

Cartograph — The Stack

Every tool the build needs, and what each one is for · Jagath Manjunath · September 2026

Your machine

WSL2 + Ubuntu	Linux on the Windows laptop. Docker needs it, Python is less painful there, and Linux fluency is on the Coverage Map anyway.
VS Code	Editor. With the WSL, Python and Docker extensions.
Git	Version control. Already installed.

Language and core

Python 3.12	The whole backend.
uv	Modern package manager. Replaces pip and virtualenv — far faster, and what current projects use.
Pydantic	Defines the shape of data and validates it. Everything in FastAPI runs on it.
pytest	Tests.

The backend

FastAPI	The web framework. The API — question in, answer and trace out. Standard for AI backends.
Uvicorn	The server that actually runs FastAPI.
SQLAlchemy	Talks to Postgres from Python without writing raw SQL for app queries.
Alembic	Database migrations — versioning the schema as it changes.

The database

Postgres 16	Two jobs: the demo warehouse being queried, and the app's own store for users, history and traces.
pgvector	Postgres extension for embeddings. Semantic search inside the database instead of a separate vector store.
Redis	Caching and rate limiting. A repeated question shouldn't cost an API call.

The AI layer

An LLM API	Anthropic or OpenAI. Writes the SQL. The only thing here that costs money.
NetworkX	Python graph library. Builds the schema graph and finds join paths between tables.
An embedding model	Turns table and column names into vectors so they can be searched semantically.

The safety layer

sqlglot	Parses SQL into a tree so it can be inspected <i>before</i> it runs — this is what rejects anything that isn't a read-only SELECT. The single most important library in the
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project.

The frontend

Node.js	Needed to run anything JavaScript.
Next.js + TypeScript	The web interface.
Tailwind	Styling without writing CSS files.
React Flow	Renders the schema graph with the chosen join path highlighted. The visual centrepiece.

Running and shipping

Docker + Compose	Packages app, Postgres and Redis so the whole thing starts with one command — laptop or server.
GitHub Actions	Runs the tests automatically on every push.
A host	Railway, Render or Fly.io — where it goes live. Free tiers exist.

Accounts needed

GitHub	Have it.
LLM API key	Costs money — budget a few hundred rupees for the whole build.
Hosting account	Free tier is enough.

Everything except the LLM API is free. And almost none of it is installed manually — Docker brings Postgres and Redis, uv brings every Python library in one command. Four manual installs: WSL, Docker, VS Code, Node. The rest arrives with the project.